REFERENCE SCENARIO / M001

Healthy machine, normal sound, non-idle

A one-machine sample based on the normal-labeled training recordings. The PDF is a readable summary; upload the WAV/CSV assets in the companion ZIP to run inference.

Machine ID	M001 - CNC Machine 01	**Example meter**	8.4 kW current / 9.2 kW baseline
Idle rule	Non-idle: 8.4 kW is above the 3.0 kW cutoff	**Factory demand**	94.6 kW / 100 kW limit

Model-scored reference example

Check	Dataset signal / condition	Example output
Machine audio	Normal MAFAULDA recording; microphone is channel 8.	**5/100 anomaly**
Vibration	Same normal recording; XYZ axes are channels 2-4.	**1/100 anomaly**
Electrical	Illustrative 8.4 kW meter reading vs 9.2 kW baseline.	**9/100 deviation**
Fused health	Prototype 40/35/25 electrical/vibration/audio score fusion.	**95/100 - Healthy**

Normal-sound reference

The supplied valve WAV is labeled normal in its source dataset. The trained valve model scored it 5/100, below the prototype 55/100 flag threshold, so the example output is "No Abnormal Sound Flagged." This is not proof that a real line is leak-free; the training labels are not leak-specific.

Upload guide

Machine Health: choose `M001_normal_reference.csv` for both the audio and vibration upload prompts. It is the original headerless 8-column, 250,000-row, 50 kHz normal recording. Audio uses column 8; vibration uses columns 2-4.

Air Leak screen: choose `M001_valve_normal_reference.wav`. A normal-labeled sound reference does not establish a leak-free condition.

Important interpretation

Outputs above were calculated from the current experimental model bundles on these exact samples. Small test sets, lab sensors, and uncalibrated model scores mean another recording or deployed model can differ. The 3.0 kW idle cutoff and meter values are illustrative app readings, not dataset labels; the idle check is rule-based, not an idle ML model.